Fast Food Chain Campaign

Akhmad Fauzi Nurulhamzah

Background 01

02 Characteristic of Store

"How is the characteristic of the store in each type of market size?"

Promotion Evaluation 03

"Which promotion/campaign brings the greatest effect on sales?"

04 Conclusion

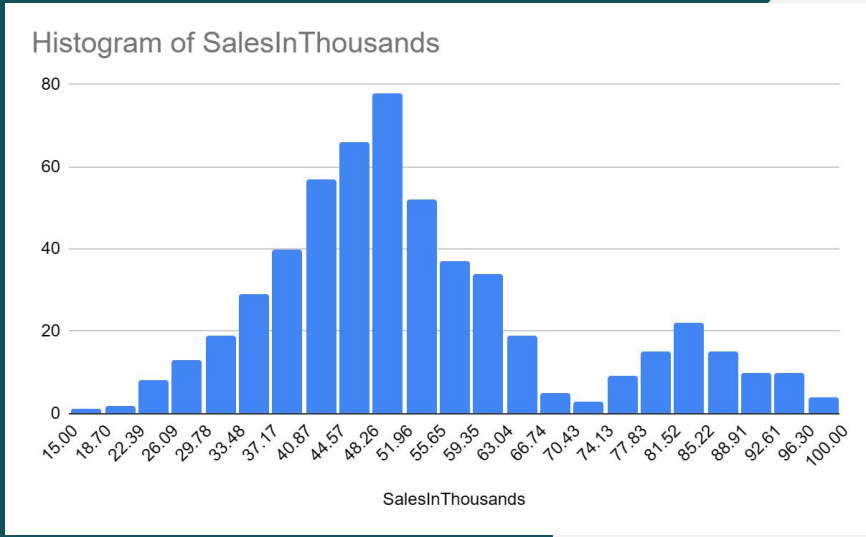
Here you could describe the topic of the section

Background

"A fast-food chain plans to add a new item to its menu. However they are still undecided between three possible marketing campaigns for promoting the new product. In order to determine which promotion has the greatest effect on sales, the new item is introduced at location in several randomly selected markets. A different promotion is used at each location, and the weekly sales of the new item are recorded for the first four weeks."

‘The company want to determine which promotion has the greatest effect on sales.”

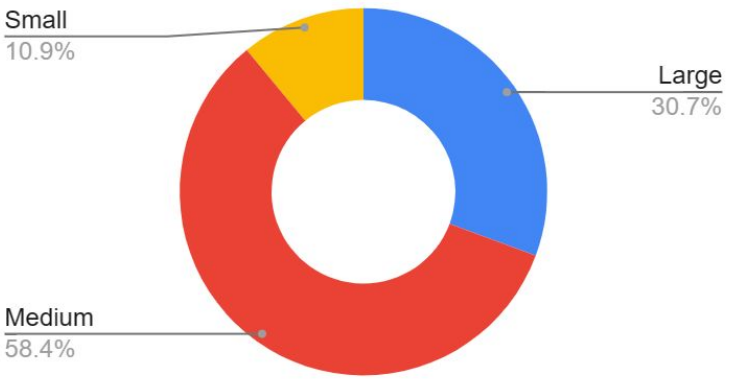
The data will help the company to find and choose right campaign by analyzing what kind of campaign that gives the most sales. We can use the columns to extract insights using statistics (Anova and Correlation) and pivot tabel/using chart



SalesInThousands	
Mean	53
Standard Error	0.7
Median	50
Mode	51
Standard Deviation	16.75521582
Skewness	0.806496413
Range	82.31
Minimum	17.34
Maximum	99.65
Sum	29299.48
Count	548
Confidence Level(95%)	1.405949831

Characteristic of Store

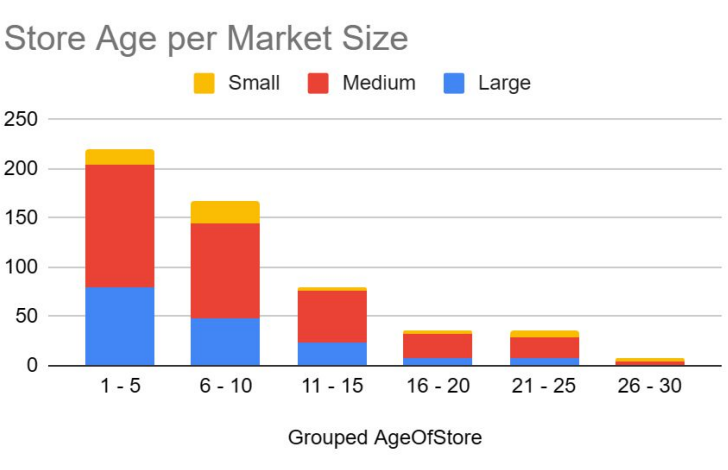
“How is the characteristic of the store in each type ofmarket size?”



More than half of the store is in the medium market, 30.7% is in the large market, and 10.9% is in the small market.

A huge market size would be more advantageous for selling the product; therefore, a data analyst might investigate the element that caused the stores to have a large market audience.

	MarketSize		
	Large	Medium	Small
COUNTA of LocationID	168	320	60

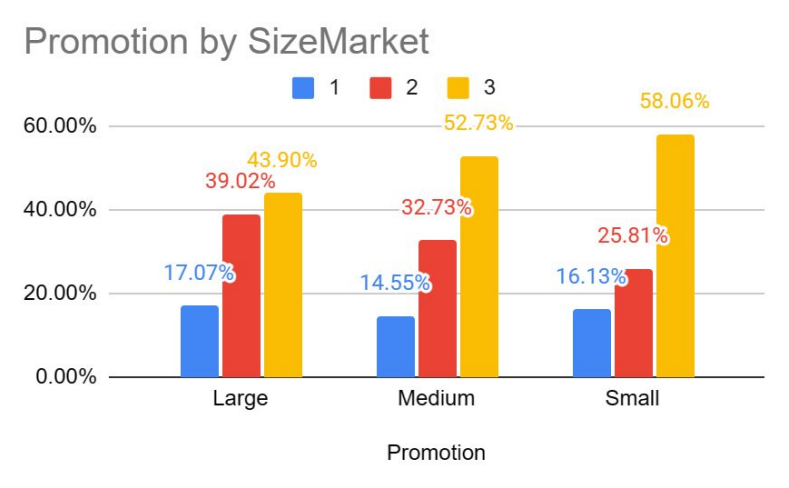


Stores under the age of 10 are the most dominant in all market sizes.

This information could be used by data analysts to search for deeper investigation into why many stores only last a short time.0 - 10 is the maximum allowed

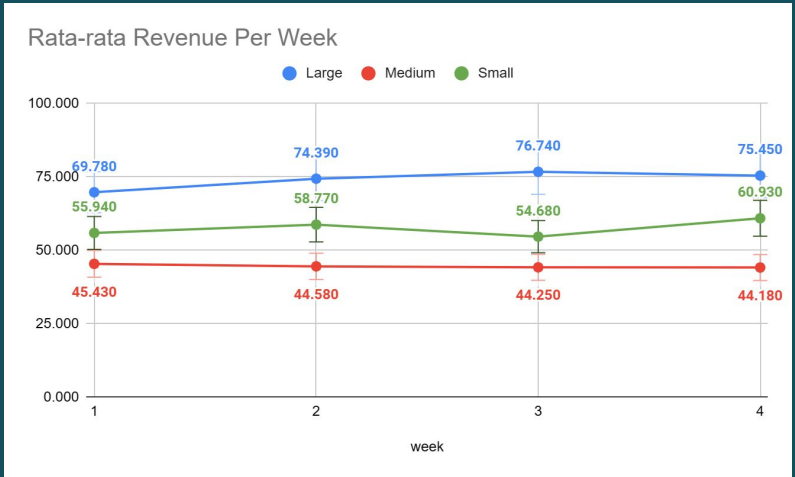
The graphs show % of promotion carried out in each market size. And the highest promotion done in every market size is promotion 3, then promotion 2, and the last is promotion 1

By this information we can asses the effectivenes for the promotin by compare it with Sales



Overall, sales in the huge market size progressively expand over the course of the provided period. While sales in small markets drop dramatically in the third week. The size of the medium market is steady.

Data analysts could go deeper into the factors that contribute to such an occurrence.



Promotion Evaluation

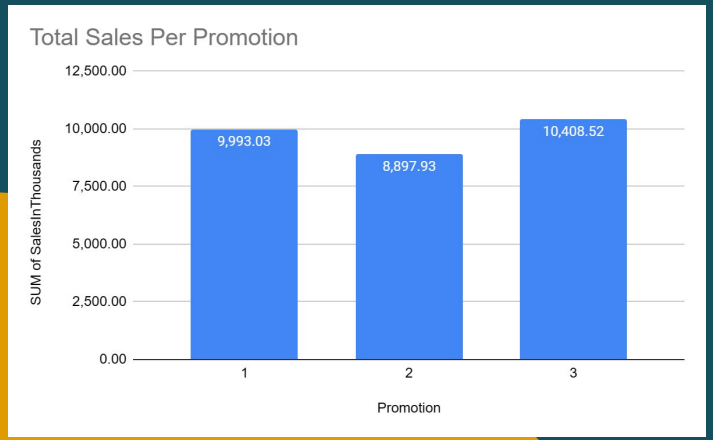
"Which promotion/campaign brings the greatest effect on sales?"

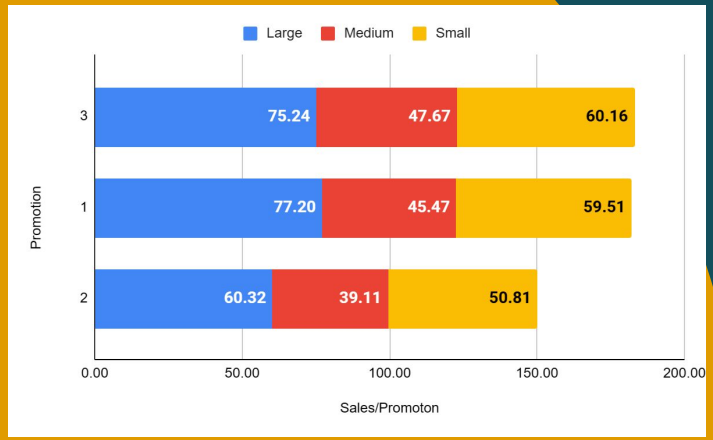
Promotion Evaluation

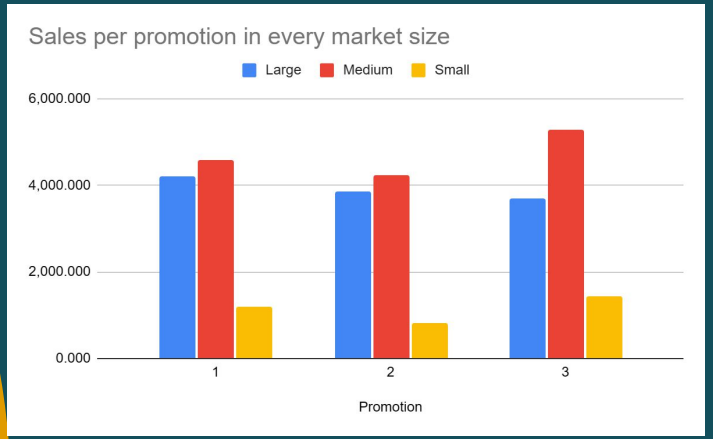
By evaluating the graph, we can see that the most promotion that giving the highest sales is promotion 3

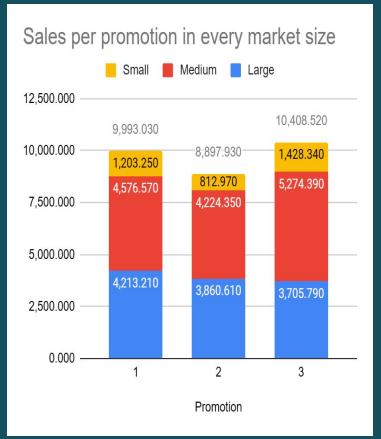
Large and medium market sizes created the most sales with the third promotion method, whereas small market sizes generated the most sales with the first promotion strategy.

By seeing the sales/promotion we see that the **promotion 3** gives the highest number of sales. which means we can assume that it is the most effective promotion eventhough we need to analyze the effectiveness in every market weather it is the right campaign in every market size









Anova: Small Market Size

Ho: μ1=μ2=μ3

Ho: each promotion has the same effect in sales in the small market size

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Promotion 1	20	1203.25	**60.1625**	26.28898816
Promotion 2	16	812.97	50.810625	34.45473958
Promotion 3	24	1428.34	59.51416667	27.17210362

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	954.6135212	2	477.3067606	16.57648113	0.000002115939996	3.158842719
Within Groups	1641.270252	57	28.79421495			
Total	2595.883773	59				

From performing the test, we have that the P-value is < threshold which is 0.05, so we can reject the Ho and there are differences effect between the promotion

And by the summary we can see that the **promotion 1** has the best resul by average in **small market size**

Anova: Medium Market Size

Ho: μ1=μ2=μ3

Ho: each promotion has the same effect in sales in the Medium market size

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Promotion 1	96	4576.57	**47.67260417**	65.08182157
Promotion 2	108	4224.35	39.11435185	77.57041733
Promotion 3	116	5274.39	45.46887931	65.47020656

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	4122.975904	2	2061.487952	29.68813375	0	3.024221928
Within Groups	22011.88146	317	69.43811185			
Total	26134.85736	319				

From performing the test, we have that the P-value is < threshold which is 0.05, so we can reject the Ho and there are differences effect between the promotion

And by the summary we can see that the **promotion 1** has the best result by average in **medium market size**

Anova: Large Market

Ho: μ1=μ2=μ3

Ho: each promotion has the same effect in sales in the Large market size

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Promotion 1	56	4213.21	75.23589286	240.1724137
Promotion 2	64	3860.61	60.32203125	247.3866355
Promotion 3	48	3705.79	**77.20395833**	207.3687606

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	10018.42096	2	5009.21048	21.44511116	0.000000005261567515	3.050787008
Within Groups	38541.17254	165	233.5828639			
Total	48559.5935	167				

From performing the test, we have that the P-value is < threshold which is 0.05, so we can reject the Ho and there are differences effect between the promotion

And by the summary we can see that the **promotion 3** has the best result by average in **Large market size**

Promotion Evaluation

"Does the type of market size affect the result of promotion?"

Check by Correlations

	MarketSize	*SalesIn*	*AgeOfStore*	*Promotion*	*week*
Market Size	1				
Sales InThousands	**0.45**	1			
Age Of Store	-0.16	-0.03	1		
Promotion	-0.06	-0.06	0.06	1	
week	0.00	-0.01	0.00	0.0	1

By seeing the table, we can see that Market size is the variable that has the big correlation with sales an it is positive correlation, thus we can assume that the bigger the market size, the sales also bigger

The correlation between market size and promotion is -0.06, and to make sure of it ststisticslly, we will perform ANOVA test

Ho: The type of market size doesn't affect the result for each campaign

From performing the test, we have that the P-value is < threshold which is 0.05, so we can reject the Ho say that **statistically the type of market size does affect the result of promotion**

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Sample	33239.26043	2	16619.63021	1261.023009	0	3.063203853
Columns	2292.031079	2	1146.01554	86.95451978	0	3.063203853
Interaction	333.5523792	4	83.38809479	**6.327114675**	**0.0001070376372**	**2.438739218**
Within	1779.230088	135	13.17948213			

Conclusion

The marketing team could consider applying promotion/campaign based on what is the most effective promotion at that market size for gaining the best result which is:
Promotion 1 in Small and Medium market size
Promotion 3 in Large Market Size

We need to analyze why many stores only last a short period of time

According to the results of the ANOVA test, it is clear that market size—small, medium, and large—had an effect on the number of sales produced by all different kinds of promotions that were used in various stores.

Thank You

